CW-ERM: Improving Autonomous Driving Planning with Closed-loop Weighted Empirical Risk Minimization

Eesha Kumar[1]*, Yiming Zhang[2], Stefano Pini[1], Simon Stent[1],
Ana Sofia Rufino Ferreira[2], Sergey Zagoruyko[1], Christian S. Perone[1]*

[1]Woven Planet United Kingdom Limited [2]Woven Planet North America, Inc. *Equal contribution
Contact: {firstname}.{lastname}@woven-planet.global

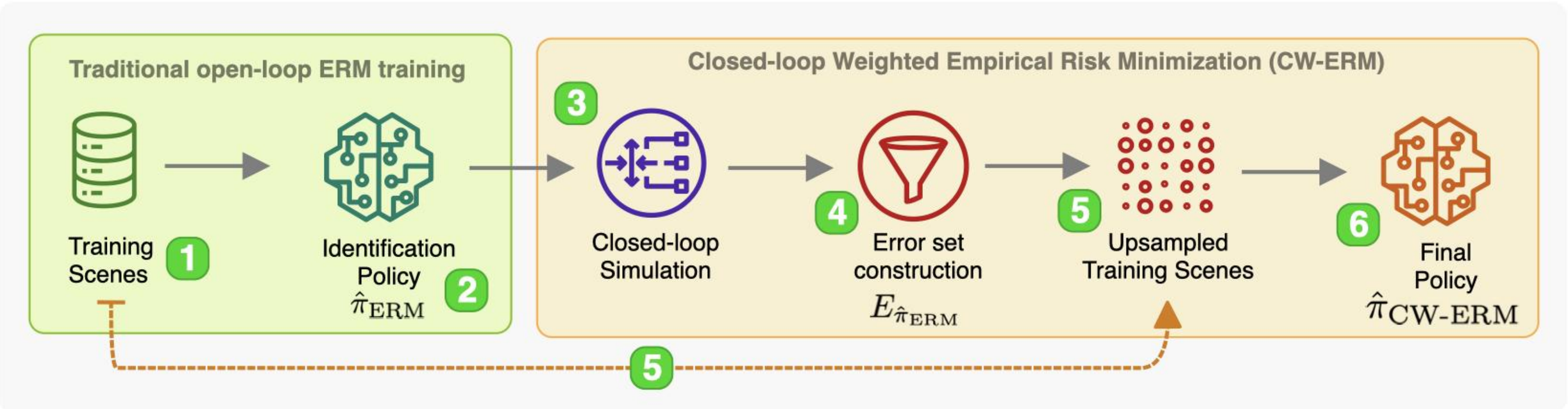
NEURAL INFORMATION PROCESSING SYSTEMS

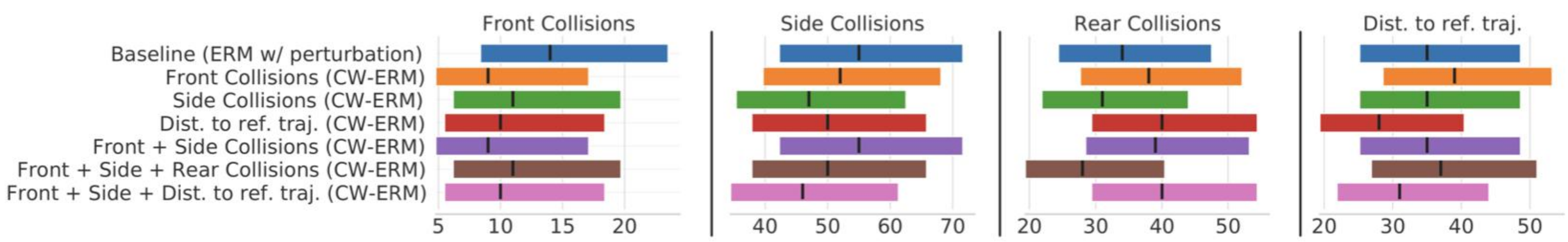
Main Ideas and Motivation

- Behavior Cloning (BC) algorithms for training self-driving vehicles via pure Empirical Risk Minimization (ERM) biases policy towards matching open-loop behavior
- We debias the policy network by identifying important samples via a closed-loop evaluator
- Training in three phases
  - Open-loop identification phase
  - Error set construction and evaluation phase
  - Upsampling training phase

Our Contributions

- Motivate and propose a technique that leverages closed-loop evaluation metrics acquired from policy rollouts in a simulator to debias the policy network and reduce the distributional differences between training (open-loop) and inference time (closed-loop);
- Evaluate CW-ERM on a challenging urban driving dataset in a closed-loop fashion to show that our method, although simple to implement, yields significant improvements in closed-loop performance without requiring complex and computationally expensive closed-loop training methods;
- Show an important connection of our method to a family of methods that addresses covariate shift through density ratio estimation.





Methodology

Stage 1 (Identification Policy)

- Train policy network via BC in open-loop using ERM, get $\hat{\pi}_{ERM}$

Stage 2 (Closed-Loop Simulation)

- Perform rollouts using $\hat{\pi}_{ERM}$ in closed-loop simulator
- Collect closed-loop metrics and identify the error set

$$E_{\hat{\pi}_{ERM}} = \{(s_i \,.\, a_i) \text{ s.t. } C(s_i, a_i) > 0\},$$

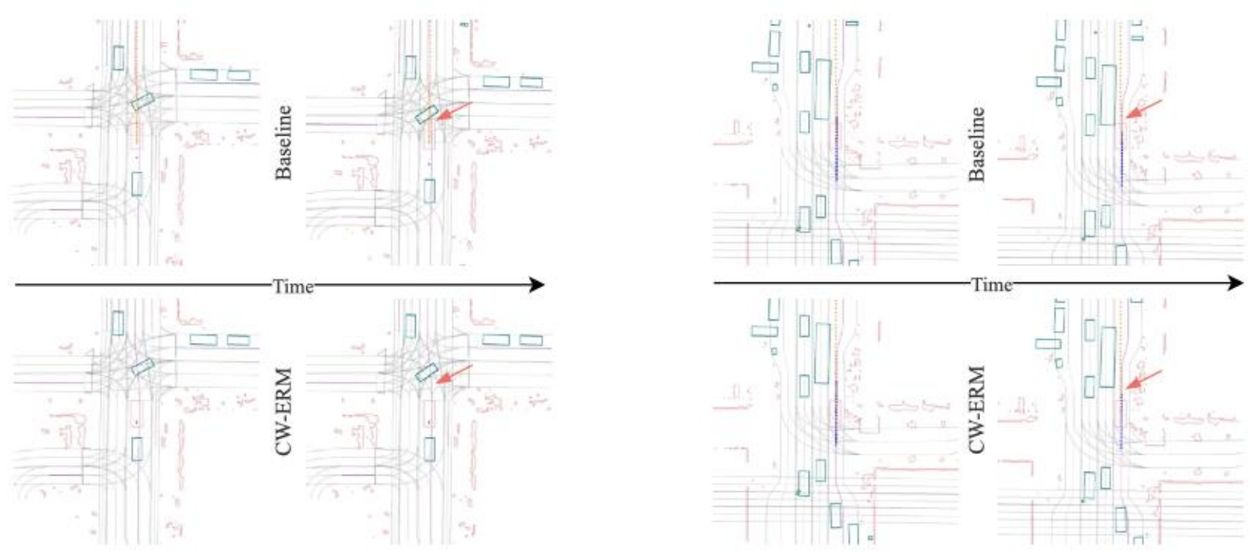
where $C(s_i, a_i)$ is a cost (e.g. collisions) incurred during the closed-loop rollouts.

Stage 3 (Final Policy)

- Train new policy where scenes belonging to $E_{\hat{\pi}_{ERM}}$ are unweighted by a factor $w(\cdot)$.

$$\arg\min_{\pi \in \Pi} \mathbb{E}_{s \sim d_{\pi^*}, a \sim \pi^*(s)} [w(E_{\hat{\pi}_{ERM}}, s) \ell(s, a, \pi)]$$

Collision Avoidance Examples



Front collisions *Side collisions*

References

1. Liu et al. (2021). Just Train Twice: Improving Group Robustness without Training Group Information. *International Conference on Machine Learning.*
2. Vitelli et al. (2022). Safetynet: Safe planning for real-world self-driving vehicles using machine-learned policies. *International Conference on Robotics and Automation (ICRA).*
3. Scheel et al. (2022). Urban driver: Learning to drive from real-world demonstrations using policy gradients. *Conference on Robot Learning.*

woven.mobi/cw-erm